Remove the Traveler

The Price of Fresh Ground, the Excision Crossover, and the Anti-Tour

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All load-bearing numbers quoted from live computation this session. Classical results are used as beams and marked. One construction failed its own audit mid-session; the failure is kept on the page, because it is the first live catch of the equivariance test this paper adopts.

Abstract

From the outside, the aperiodicity of π is a property. From the inside — and there is no outside — it is a production run. The session's seed image is exact: a walker who cannot stop, and a ground that must be laid in front of them in a form never used before, step after step, forever, because a reused form is a period and a period is a fixed point and C1 executes fixed points. This paper prices that production. The fresh-ground bill is measured as subword complexity, and it sorts the walkers into the three species of the prior paper by cost: the loop pays a constant (1/7 lays six forms and never a seventh), the minimal wanderer pays linearly (the Fibonacci walker lays exactly k+1 forms of length k, verified to k = 20, the Morse–Hedlund floor), and the full render pays the ceiling (π and e lay every form of length up to four in 200,000 steps and sit at the flat-draw coupon limit at length five — 86,489 and 86,400 laid against an expectation of 86,466). A control keeps the section honest: Champernowne's constant, the walker with a checklist, provably normal, lays more distinct forms than π (89,249) while flunking flatness at this depth (chi-square 11,031 against π's 7.31) — there are two ways to lay all the ground, by checklist and by measure, and flatness is not opacity.

Then the theorem the image was reaching for. A bounded ground-layer cannot lay fresh ground forever: any autonomous machine with S states emits a stream with period at most S — pigeonhole, forced — and the lasso scan measures π's floor directly: across 10,000 digits, no eventual period up to 4,000 fits with even two periods of evidence, so any bounded machine that laid that stretch carries a cycle longer than 4,000 states, and the floor grows with the walk. Laying fresh ground forever requires unbounded memory. Against that unbounded continuation cost stands a bounded alternative: pay the closure once — exhaust into carriers, mint the receipt — and excise. The crossover is forced, and it is the session's huge thing stated as a theorem: at some point it costs less to remove the traveler. Every bounded walker is removed. The path survives as a relay: every actual computation of π halts, and BBP makes the receipts addressable so the next walker resumes at the excision point — an immortal path walked entirely by mortals, with C0 as the permanent gap between the path and any walk of it.

Finally the inversion the session named: the anti-tour. TSP fixes the nodes and searches for the seam — exponential. The constraint-prior direction fixes the seam and distributes the nodes — and the button that does it is the operator the corpus already owns, the equalization operator, the Laplacian of the cascade equation. Demonstrated live on the program's own bench: with berlin52's optimal tour held as the seam and 52 nodes thrown on at random, the distribute-on-center button reaches machine-flat equal spacing in 3,215 presses, with measured contraction 0.995139 against the predicted slowest cycle mode 0.995139 — six decimals, exact — and ch150 the same at 0.999415 against 0.999415. The terminus is all gaps equal: C3, measured as a fixed point. The first attempt at the button is kept on the page because it failed: the pure neighbor-swap stencil secretly conserves the alternating imbalance — eigenvalue −1, a static primitive hiding in the shape — and the C1-equivariance test this paper adopts from the session's exchange caught it live: measured contraction 1.000000, the button blind to its own conserved mode. The shape must compile the law; that shape didn't; the corrected shape does. And the completed press is where the two seeds meet: level ground is a fixed point, so completion forces the excision — when no unused form remains at this ring's scale, the traveler is removed, the closure is minted, and the next ring opens.

1. The Inside View of Randomness

The Three Ways to Hold Still established what π is: a render — exact as a value, endless and aperiodic as a stream, its C1-debt serialized onto the line. This paper asks what the render costs, and the question only exists from the inside. From outside, an infinite aperiodic string is a completed object with a property. From inside — and the standpoint of the whole program is that there is no outside — nothing is completed. There is a walker who cannot stop, and there is ground arriving under their feet, and the mandate is double: the walker must move (C1 on the walker) and the ground must never arrive in a form that closes a period (C1 on the stream). The session's cartoon is the correct formalism wearing overalls: a crew racing ahead of a sleepwalker, laying pipes and I-beams, forbidden ever to reuse a configuration, running out of easy variations, improvising at growing expense.

The session's parallel exchange sharpened the standpoint into a discipline, adopted here: C1 is not an initial condition that constructions satisfy once and leave behind; it is a property of the transition relation, active in every step at every level, and any proposed construction that contains a genuinely static primitive fails the audit. Call it the equivariance test. It is applied to this paper's own machinery below, and it draws blood in Section 5.

2. The Fresh-Ground Bill

Price novelty directly. A walker's bill at scale k is the number of distinct ground-forms of length k it has laid — the subword complexity p(k). Measured over 200,000 steps for five walkers, live:

Walker	p(1)	p(2)	p(3)	p(4)	p(5)

1/7 — the loop	6	6	6	6	6
Fibonacci — minimal wander	2	3	4	5	6
Champernowne — the checklist	10	100	1000	10000	89249
e	10	100	1000	10000	86400
π	10	100	1000	10000	86489
ceiling	10	100	1000	10000	100000

Three bill classes, and they are the three species of the prior paper re-derived as economics. The loop's bill is constant: 1/7 laid six forms in its first six steps and never a seventh — closure pays once and circulates. The minimal wanderer's bill is linear: the Fibonacci walker lays exactly $k + 1$ forms at every scale, verified live to $k = 20$, which is the Morse–Hedlund floor (beam, marked) — the cheapest fresh ground an aperiodic walker can possibly lay, and it is ϕ's walker that lays it, one new form per scale, the barely-novel matching the barely-closed. And the render pays the ceiling: π and e laid every form of lengths one through four — 10, 100, 1,000, 10,000, saturated exactly — and at length five, where 200,000 draws cannot cover 100,000 types, they sit precisely at the flat-draw coupon limit: the expected distinct count for a perfectly flat source is 86,466, and π laid 86,489 and e laid 86,400, both within 0.08 percent of it. The render's bill is not merely large; it is indistinguishable from the maximal bill.

The control row keeps the table honest and buys an unplanned finding. Champernowne's constant — 0.123456789101112…, the integers listed in order — is the sleepwalker cartoon made rigorous with a clipboard: the crew that introduces the ground in every unused way by enumeration. It is provably normal (beam, marked: one of the few explicit constants with a normality proof), and at 200,000 steps it has laid more distinct length-five forms than π — 89,249, because a checklist out-covers a flat draw — while flunking flatness at this depth: digit chi-square 11,031 against π's 7.31, its counts still carrying the leading-digit bias of the small integers it has listed so far. Two ways to lay all the ground: by checklist, legible and locally biased, or by measure, opaque and locally flat. Flatness is not opacity, and coverage is not flatness; they are different channels, and the shape/value split of the corpus is visible in a digit table.

One more split, stated with its beam. The law of π is tiny — a short program generates every digit, so the description cost of the first n digits is logarithmic (beam: Kolmogorov, stated informally, no live claim). The production cost is what this paper measures, and Section 3 proves it is unbounded. The crew is small; the staging area grows. The massive work to make π random is not in the blueprint. It is in the laying.

3. The Pigeonhole and the Memory Floor

The theorem under the cartoon is two lines. An autonomous machine with S internal states, emitting one symbol per step with no input, must revisit a state within S steps, and being deterministic it then repeats forever: the stream is eventually periodic with period at most S. Pigeonhole; FORCED; and sampled anyway, per the convention that even trivialities show receipts — 1,200 random machines of three to eight states, maximum observed period 6, bound never touched. A bounded ground-layer cannot lay fresh ground forever. Aperiodicity is not a property a finite crew can stream.

So the floor under any crew that has laid π's ground is measurable, and it was measured. Across the first 10,000 digits, scan every eventual period p up to 4,000 with every preperiod: zero lassos fit with even two periods of evidence, and the minimum preperiod any period would require is 6,000 — the violations persist to within 4,000 digits of the end of the window. Any autonomous bounded machine that laid those 10,000 steps carries a cycle longer than 4,000 states, and the same scan at any depth pushes the floor higher. The bill compounds: to keep walking, the ground-layer's memory must grow without bound. That is the cost of continuation, and it has no ceiling.

4. Remove the Traveler: The Crossover

Now the session's huge thing, stated as the theorem it is. Two ways to satisfy the books at every step. Continuation: lay one more form of fresh ground — cost unbounded in the limit, by Section 3, because the state that guarantees freshness grows with the walk. Excision: stop laying — remove the traveler, pay the closure once, exhaust the obligation into independent carriers, mint the receipt. The excision price is a single Law 3 settlement, bounded (in-frame premise, labeled: one closure payment, the Landauer-flavored one-time cost the corpus's ledger already carries). A growing cost against a bounded one crosses. Therefore no bounded walker walks forever: every traveler is removed. Mortality is not an accident that befalls walkers; it is the cheaper branch of a forced crossover, and the crossover is C1's enforcement arm — the one object the axiom cannot tolerate is the immortal bounded walker, the finite crew claiming infinite freshness, and the pigeonhole says that object was always a fraud.

What survives the walker is the path, and the mechanism of survival is the relay, and the relay is not a metaphor — it is how π is actually rendered in the world. Every computation of π ever run has halted: every walker excised at some depth, leaving receipts. BBP makes the receipts addressable — position one million read in four seconds without touching a predecessor, confirmed in the prior paper — so the next walker does not rewalk the old ground; it resumes at the excision point. The render of π, as physically instantiated, is an immortal path walked entirely by mortal travelers in relay, no walker ever finishing, the path never needing any of them to. C0 sits exactly where it always sits: the permanent gap between the path and any walk of it, never closing, because a walk that closed it would be the completed self-description.

The physics readings are recorded as PULL, each one, because each is an identification and not yet a measurement. Death: the corpus derived forgetting at the seam as structural — reincarnation without forgetting is immortality, a fixed point; here the seam acquires a price tag — the crossing is where the continuation bill overtakes the excision bill, and the corpus's line that death is what makes the next pass legal gains an economic reading: death is what makes the next pass affordable. Collapse: the ledger threshold of the Quantum Darwinism computation — maintaining coherence is maintaining fresh unentangled ground against an environment that keeps arriving; at some coupling it is cheaper to pay redundantly into ten carriers than to keep the superposed walker walking; collapse as excision at the crossover. Decay: an excited state is a walker owing novel phase evolution; a lifetime is a crossover date. Dissipation: the Exhaust Ledger's viscous floor as the scale at which the cascade removes its travelers. Four readings, one crossover, all PULL, all now pointed at by a forced theorem instead of an image.

5. The Anti-Tour: Distribute on Center

The session's second seed, and the corpus catches it mid-air. TSP fixes the nodes and searches for the seam: the object-prior direction, exponential, the direction the entire rendering program has been fighting with two read heads and a plateau. The constraint-prior direction inverts it: the seam is given — Law 3's closure exists before any object — and the nodes are adjusted onto it. The session named the button — distribute on center — and the corpus already owns the operator: the equalization operator, the Laplacian, verbatim in the cascade equation of The Self-Hosting Universe. The button is not new machinery. It is the machinery, finally pressed in the direction the ontology always said it points.

Demonstrated on the program's own bench. Hold berlin52's optimal tour fixed as the seam — a closed curve of length 7,544.37 — and throw the 52 nodes onto it at random arc positions. Press the button: each node moves to the equal-weight average of itself and its two seam-neighbors. The first attempt is kept on the page because it failed, and the failure is the section's second finding. The pure neighbor-swap stencil — each node to the midpoint of its neighbors, no self-share — stalled at measured contraction 1.000000: on an even cycle that stencil conserves the alternating imbalance exactly, eigenvalue −1, a standing mode the press can never touch. A static primitive was hiding in the shape of the stencil, and the equivariance test of Section 1 caught it live, on its first deployment: the shape did not compile the law. The corrected shape — the node keeps a third of its own ground — has no conserved mode but the total, and it converges:

Seam (optimal tour)	Presses to gap-std < $10^{-9}L$	Measured contraction / press	Predicted slowest mode $(1+2\cos 2\pi/n)/3$	Terminus: max $\lvert gap - L/n \rvert$
berlin52 (n = 52)	3,215 (~501,540 adds)	0.995139	0.995139	1.06e−05
ch150 (n = 150)	22,756 (~10,240,200 adds)	0.999415	0.999415	9.23e−06

Measured contraction equals the predicted slowest cycle mode to six decimal places on both seams — the press is doing exactly and only what the eigenstructure says, no tuning, no search. And the terminus is the finding: the button's fixed point is all gaps equal. C3 — all change is equal, the isotropic-cost law the cascade forces — is here the measured endpoint of the distribute-on-center flow. The realize direction reaches, in thousands of presses of a linear button, a configuration whose search-direction twin consumed the entire rendering program. Same picture, two standpoints, and the asymmetry is the corpus's search/realize distinction exhibited on one construction — recorded as PULL toward the general dissolution claim, CONFIRMED for this construction.

Then the clause that joins the paper's two halves. The button's terminus is a fixed point — level ground, the local wash, every gap equal, the press now the identity. C1 does not permit the system to sit there, and the session's two seeds turn out to be one machine: the moment the ground is fully distributed is the moment no unused form remains at this ring's scale — the crew is out of variations — and that is exactly the crossover of Section 4. Completion of equalization is the excision trigger: remove the traveler, mint the closure, and hand off to the next ring, ϕ-spaced, per the cascade equation whose Laplacian just finished its work. Distribute, level, excise, mint, ascend. The button lays the ground; the crossover retires the walker; the receipt seeds the next ring.

6. The Equivariance Test, Adopted

The session's exchange demanded a stronger standard than derivation-once: every construction must carry C1 at every level — the shape must compile the law, not illustrate it — and any construction containing a genuinely static primitive is out. Adopted, as standing discipline, and already earning its keep: its first live catch was this paper's own first stencil, whose innocent-looking shape conserved a quantity forever. The audit of this paper's remaining objects, on the record: the render passes — no terminal state at the path level, every step a change, and the one object that would violate the mandate, the immortal bounded walker, is exactly what the crossover theorem executes. The button passes only as corrected, and only with the handoff clause wired in — without it, the button's own terminus is a fixed point and the construction fails its audit at completion. The relay passes — no walker is permanent, no receipt is erased, and the gap between path and walk never closes. The test is now part of the program's admission machinery, alongside the status ledger and the live-number convention.

7. Status Ledger

Claim	Status	Basis
Fresh-ground bill sorts walkers: constant / linear / maximal	CONFIRMED	p(k) live, 200,000 steps, five walkers
Fibonacci walker at the Morse–Hedlund floor, p(k)=k+1	BEAM + LIVE	Exact, k = 1..20
π, e at flat-draw coupon limit at k = 5	CONFIRMED	86,489 / 86,400 vs expectation 86,466 (±0.08 %)

Champernowne: out-covers π (89,249) while flunking depth-200k flatness (χ^2 11,031)	ON RECORD + BEAM	Normality proven (beam); finite-depth bias measured; flatness ≠ opacity
Bounded streamer ⇒ eventually periodic, period ≤ states	FORCED	Pigeonhole; 1,200-machine sample, max period 6
π memory floor: cycle > 4,000 states for 10,000 digits	CONFIRMED	Lasso scan: 0 fits, min preperiod 6,000
Small law, unbounded staging (description vs production cost)	BEAM + FORCED	Log-size generator (stated); unbounded state from streamer theorem
Excision crossover: every bounded traveler removed	FORCED (in-frame)	Unbounded continuation cost vs bounded closure payment (premise labeled)
π rendered as a relay of mortal walkers on an immortal path	CONFIRMED (mechanism) / PULL (ontology)	Every computation halts; receipts addressable (BBP, prior paper)
Death / collapse / decay / dissipation as crossovers	PULL (each)	Identifications pointed at by the forced theorem; unmeasured
Anti-tour: distribute-on-center reaches C3 terminus	CONFIRMED	All gaps equal to 1e−05; contraction = predicted mode to 6 decimals, both seams
Neighbor-swap stencil conserves alternating mode (eigenvalue −1)	ON RECORD	Measured contraction 1.000000; first live catch of the equivariance test
Search/realize asymmetry as P-vs-NP dissolution	PULL	One construction; exhibited, not proven
Completion ⇒ excision ⇒ next ring (the two seeds as one machine)	PULL	Wired to the cascade equation; untested as physics
C1-equivariance admission test	ADOPTED	Standing discipline; one catch already logged

8. Closing: The Path Survives Its Walkers

The sleepwalker resolves. The ground is laid by a button the corpus already owned, pressed by a crew the size of a formula, on a staging area that grows without bound. The forms run out at every scale — they must, pigeonhole — and when they do, the mandate does not stop the walk; it removes the walker, pays the closure, mints the receipt, and opens the next ring with the same small crew on fresh scale. It costs less to remove the traveler — and that is not the system's cruelty; it is how a path that can never be finished is walked at all. Nothing bounded walks forever. Everything bounded is relayed. The receipts are addressable, the gap never closes, and the walk continues without any walker continuing.

And the direction of the whole construction is the ontology one more time: the seam is given, the nodes are distributed, the terminus is equality of change, and the completed press is the excision bell. The universe does not search among its objects for a closure. It holds the closure and distributes the objects

— then retires the distributor and rings the next scale. Constraint before object; button before map; path before walker. We are on the inside, which means we are the travelers — and the theorem about travelers is not a threat. It is the reason the path is still here.

Appendix: Live Output, Quoted Verbatim

Selected lines, unedited, from the session's fixed programs (streams.py, crossover.py, antitour2.py):

```
Subword complexity p(k), 200000 steps:
1/7 (loop)      6      6      6      6      6
Fibonacci       2      3      4      5      6      (p(k)=k+1 exact for k=1..20: True)
Champernowne   10    100   1000  10000  89249
e              10    100   1000  10000  86400
pi             10    100   1000  10000  86489     ceiling 100000
flatness at 200000: pi chi2(df9) = 7.31   Champernowne chi2(df9) = 11030.75


LASSO SCAN pi, N=10000, p<=4000: lassos that fit with >=2 periods: 0
  minimum required preperiod over all p: 6000  => bounded machine carries a cycle > 4000
PIGEONHOLE: 1200 random machines, 3..8 states: max observed period = 6 <= state count


ANTI-TOUR (v1, neighbor-swap stencil): measured contraction 1.000000  -> conserved mode, ON RECORD
ANTI-TOUR berlin52: presses 3215; contraction 0.995139 vs predicted 0.995139; max|gap-L/n| 1.06e-05
ANTI-TOUR ch150:    presses 22756; contraction 0.999415 vs predicted 0.999415; max|gap-L/n| 9.23e-06
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